

Object class

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By default all the java classes inherits Object class,if explicitly one class is not inheriting other class.

```
class Animal // Super class of Animal is Object
{
}

class Dog extends Animal // Super class of Dog is Animal
{
}
```

Note : **Interface** not inherits Object class by default

Object class toString method

class Employee{...}
Super class of Employee is Object

```
public String toString() {
    return getClass().getName() + "@" + Integer.toHexString(hashCode());
  }
```

class Employee //extends Object → added by default
{
    Object class toString method is inherited
}

Object class toString method

Before overriding toString method

Printing reference variable - internally java calls the inherited toString method / Object class toString method that's why prints the reference.

After overriding toString method

Printing reference variable – calls the overridden toString method

Object class toString method

```
String name = “Abi”;
System.out.println(name); // Abi
```

Super class of String is Object and String class overrides the toString method of the Object class and logic is already written to print the string.

Object class equals method

Object class equals method :

```
public boolean equals(Object obj) {
    return (this == obj);
  }
```

```
Employee employee1 = new Employee(102, "John"); //123
Employee employee2 = new Employee(102, "John");//456
employee1.equals(employee2)//false
```

Object class equals method compares the references of two employee objects.

```
String name1 = "sachin";
String name2 = "sachin";
System.out.println(name1.equals(name2)); // true
```

By default String class overrides the equals method of the Object class and logic is written to check the content of the two string objects.

Object class equals method

employee1.equals(employee2));//false

By default, calls the Object class equals method.

After overriding the equals method
employee1.equals(employee2);//true

calls the overridden equals method

Object class equals method

Without overriding the equals method

employee1.equals(employee2)
calls the Object class equals method / calls the inherited equals method

After overriding the equals method

employee1.equals(employee2)
calls the overridden equals method

THANK YOU